ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 2

INTRODUCTION TO SYSTEMS DEVELOPMENT

PAPER 1 · PRACTICE A

MARKS: 100 **TIME: 3 hours**

NC(V) Information Technology and Computer Science, Level 2. Section A: objective questions (30) · Section B: the systems development life cycle, problem solving and algorithms, pseudocode and flowcharts, variables, data types and control structures, testing (70).

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 2. It is not a Department of Higher Education and Training examination paper.

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Algorithms may be written in pseudocode or drawn as flowcharts, unless a specific method is asked for.
5. Write neatly and legibly.

SECTION A

QUESTION 1

1.1 Various options are given as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question number (1.1.1–1.1.10).

1.1.1 A step-by-step set of instructions to solve a problem is called a/an …

A variable. B compiler. C interface. D algorithm.

1.1.2 In a flowchart, a decision is shown by a …

A diamond. B rectangle. C parallelogram. D oval.

1.1.3 Which data type is the most suitable to store a learner's surname?

A Integer B Boolean C String D Real

1.1.4 The phase of the SDLC in which the new system is installed and users start using it is …

A analysis. B design.
C implementation. D planning.

1.1.5 Which operator means "not equal to" in most pseudocode?

A <> B = C >= D :=

1.1.6 A loop that is controlled by a counter and repeats a fixed number of times is a …

A WHILE loop. B REPEAT … UNTIL loop.
C IF statement. D FOR loop.

1.1.7 An error that causes a program to give wrong results without crashing is a …

A syntax error. B logic error.
C runtime error. D spelling error.

1.1.8 The result of 17 MOD 5 is …

A 3. B 3,4. C 12. D 2.

1.1.9 A Boolean variable can store …

A any whole number. B only TRUE or FALSE.
C one character. D a date.

1.1.10 The person who studies a business problem and designs a computer solution is a …

A network technician. B data capturer.
C systems analyst. D end user.

(10)

1.2 Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'true' or 'false' next to the question number (1.2.1–1.2.10).

1.2.1 A constant's value can change while the program is running.

1.2.2 The SDLC is a structured approach to developing information systems.

1.2.3 In a flowchart, a parallelogram represents input or output.

1.2.4 A WHILE loop always runs at least once.

1.2.5 The expression (5 > 3) AND (2 > 4) is FALSE.

1.2.6 A feasibility study determines whether a proposed system is worth developing.

1.2.7 Pseudocode must follow the exact syntax of one programming language.

1.2.8 Testing should only take place after the system has been handed over to the users.

1.2.9 A trace table is used to follow the values of variables step by step.

1.2.10 Integer division of 9 by 2 (9 DIV 2) gives 4.

(10)

1.3 Choose a term from COLUMN B that matches a description in COLUMN A. Write only the letter (A–J) next to the question number (1.3.1–1.3.10).

	COLUMN A		**COLUMN B**
1.3.1	Named storage location whose value can change	A	comment
1.3.2	Repeating a set of instructions	B	fact-finding
1.3.3	Choosing between two or more paths	C	terminator
1.3.4	Instructions carried out one after the other	D	iteration
1.3.5	Translates a whole program into machine code before it runs	E	compiler
1.3.6	Data that is entered into the program to check how it reacts	F	variable
1.3.7	Gathering facts with a questionnaire or interview	G	sequence
1.3.8	Shows the start or end of a flowchart	H	test data
1.3.9	Explains the program to other programmers inside the code	I	debugging
1.3.10	Fixing errors in a program	J	selection

(10)

TOTAL SECTION A: [30]

SECTION B

QUESTION 2: THE SYSTEMS DEVELOPMENT LIFE CYCLE

2.1 Name the FIVE main phases of the systems development life cycle (SDLC) in the correct order. (5)

2.2 A taxi association wants to replace its paper-based record of daily takings with a computer system. Explain what will be done in the analysis phase for this project. (3)

2.3 Name TWO fact-finding techniques and give ONE advantage of each. (4)

2.4 Explain the difference between direct changeover and parallel running as methods of converting to a new system. Which method would you recommend for the taxi association? Give a reason. (5)

2.5 Give THREE reasons why systems need maintenance after they have been implemented. (3)

[20]

QUESTION 3: PROBLEM SOLVING AND ALGORITHMS

3.1 The college canteen gives a 10% discount when a student buys for more than R50. Write an algorithm in pseudocode that asks for the total amount, calculates the discount (if any) and displays the amount to pay. (6)

3.2 Draw a flowchart for the algorithm in QUESTION 3.1. Use the correct flowchart symbols. (6)

3.3 Identify the input, processing and output for QUESTION 3.1 (an IPO chart). (3)

[15]

QUESTION 4: VARIABLES, OPERATORS AND CONTROL STRUCTURES

4.1 Suggest a suitable variable name and data type for EACH of the following: (a) the number of students in a class; (b) the price of a loaf of bread; (c) whether a student has paid the registration fee. (6)

4.2 Give the value of X after each statement. Assume A = 7 and B = 3. (a) X ← A + B * 2 (b) X ← (A + B) / 2 (c) X ← A MOD B (d) X ← A DIV B (4)

4.3 Study the pseudocode below and answer the questions.

```
1  total ← 0
2  count ← 0
3  FOR i ← 1 TO 4
4     INPUT mark
5     IF mark >= 50 THEN
6        count ← count + 1
7     ENDIF
8     total ← total + mark
9  NEXT i
10 average ← total / 4
11 OUTPUT average, count
```

4.3.1 Complete a trace table for the input values 60, 35, 80 and 45. Use the columns: i, mark, count, total. (6)

4.3.2 What will be displayed on line 11? (2)

4.3.3 What does the variable count represent? (1)

4.3.4 Rewrite lines 3 to 9 using a WHILE loop. (4)

4.3.5 Identify ONE example of selection and ONE example of iteration in the pseudocode by giving the line numbers. (2)

[25]

QUESTION 5: TESTING AND DOCUMENTATION

5.1 A program must accept a student's age between 15 and 35 (inclusive). Give ONE example each of normal, boundary (extreme) and erroneous (abnormal) test data. (3)

5.2 Distinguish between a syntax error and a logic error. Give an example of each. (4)

5.3 Name TWO types of documentation produced during systems development and state who uses each. (3)

[10]

TOTAL SECTION B: [70]

TOTAL: 100

END OF QUESTION PAPER